

Music Genre Classification

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1. Introduction

The goal of this project was to build a system that can automatically classify music into different genres using classical machine learning methods. Music genre classification is important for music recommendation, playlist creation, and organizing large audio libraries. Even though deep learning is widely used today, traditional machine learning approaches can still achieve strong performance when combined with useful audio features.

In this project, I used the GTZAN dataset, which contains ten music genres including classical, jazz, metal, and rock. I extracted meaningful audio features from each track and trained several supervised learning models to compare their performance. I also experimented with unsupervised clustering and simple generative models to understand how well the dataset can be separated without labels.

The main objectives of the project were:

- Extract useful features that represent the musical qualities of each audio clip.
- Train and evaluate multiple machine learning models for genre classification.
- Explore unsupervised and generative models for additional insight.
- Discuss results and identify potential improvements.

This work revisits ideas from prior studies such as Tzanetakis and Cook [1], who showed that MFCC-based features and classical classifiers can distinguish genres effectively.

2. Problem Statement, Setup, and Methods

A. Problem Statement

The task is to predict the genre of a music track using a machine learning model trained on engineered audio features.

B. Dataset and Setup

The project uses the GTZAN dataset, which contains 1000 tracks (10 genres, 100 tracks each). One file was corrupted and removed, leaving 999 usable tracks. Each audio file is 30 seconds long and was processed at a sampling rate of 22,050 Hz.

The dataset was split into:

- 80% training data
- 20% testing data

A stratified train–test split ensured that each genre was represented proportionally.

C. Feature Extraction

Using the Librosa library, I extracted several features that capture timbre, harmony, and spectral properties of each audio signal:

- 20 Mel-Frequency Cepstral Coefficients (MFCCs)
- 12-dimensional chroma features
- Spectral contrast
- Tonnetz harmonic features
- Spectral centroid, bandwidth, and rolloff
- Root-mean-square (RMS) energy

For each feature type, the mean and standard deviation were computed across time, resulting in a final feature vector of 157 values per track.

D. Supervised Learning Models

I trained and evaluated the following models:

- Logistic Regression
- k-Nearest Neighbors (kNN)
- Linear SVM
- RBF-kernel SVM
- Decision Tree
- Random Forest
- AdaBoost

Hyperparameters were tuned using GridSearchCV with 5-fold stratified cross-validation.

E. Unsupervised and Generative Models

To explore the structure of the dataset beyond supervised learning, I also tested:

- K-Means clustering (10 clusters)
- Bayesian Gaussian Mixture Model (BGMM)
- Gaussian Naive Bayes
- Class-conditional Gaussian Mixture Models (GMMs)

Clustering quality was measured using normalized mutual information (NMI) and adjusted Rand index (ARI).

3. Results

A. Supervised Learning Performance

The Random Forest model achieved the best performance with:

- **78.5% test accuracy**
- **0.779 macro F1-score**

Other models performed as follows:

- Logistic Regression: 73% accuracy
- Linear SVM: 71.5% accuracy
- RBF SVM: 71% accuracy
- kNN: 64% accuracy
- Decision Tree: 56% accuracy
- AdaBoost: 46% accuracy

These results show that Random Forest and SVM-based methods are strong choices for this feature space.

B. Unsupervised Learning

K-Means clustering achieved:

- $\text{NMI} = 0.160$
- $\text{ARI} = 0.083$

Similarly, the Bayesian GMM achieved:

- $\text{NMI} = 0.162$
- $\text{ARI} = 0.084$

These low values show that music genres do not naturally form clean clusters in the engineered feature space.

C. Generative Models

- GaussianNB reached a resubstitution accuracy of 47.9%.
- Class-conditional GMMs reached approximately 99.9% resubstitution accuracy, indicating heavy overfitting.

4. Insights and Future Work

A. Insights

- Classical ML methods can perform well on audio classification when strong features are used.
- Random Forest outperformed all other models due to its ability to capture nonlinear relationships and handle many features.
- Some genres (classical, metal, jazz) are more distinct and easier to classify.
- Rock, blues, and reggae overlap more and often confuse the models.
- Unsupervised methods performed poorly, showing that genre is a human-defined label, not a natural cluster.

B. Future Work

Potential improvements include:

- Training convolutional neural networks (CNNs) on mel-spectrogram images.
- Using data augmentation (pitch shift, time stretch).
- Expanding the dataset for better generalization.
- Applying sequence models such as CRNNs or Transformers for capturing temporal structure.

5. Conclusion

This project implemented a complete music genre classification pipeline using classical machine learning techniques and audio feature engineering. The Random Forest model achieved the highest performance with 78.5% accuracy. Unsupervised and generative analyses offered additional insight into the dataset’s structure and limitations. Overall, this project demonstrates that classical ML remains effective for audio classification tasks and provides a strong foundation for future deep learning work.

References

- [1] G. Tzanetakis and P. Cook, “Musical genre classification of audio signals,” *IEEE Transactions on Speech and Audio Processing*, vol. 10, no. 5, pp. 293–302, 2002.
- [2] B. McFee *et al.*, “librosa: Audio and music signal analysis in Python,” *Proc. 14th Python in Science Conference*, pp. 18–25, 2015.